Assignment 2 - Product Requirement Document - Alcove Companion AI - Saurav Dasgupta C52

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Alcove — Product Requirements Document

MVP | Product, Design & Engineering Handoff

Product: Alcove

Positioning: *Someone who remembers you.*

Vision: An emotionally intelligent AI that grows with you through every stage of life.

MVP Persona: Young professionals who live independently and want an AI companion that remembers their story, goals, and meaningful moments.

PAGE 1 — PRODUCT DEFINITION & MVP

1. Product Definition

Alcove is a **persistent relationship system** that combines conversation, memory, emotional understanding, personalization, reflection, and growth to create a feeling of being genuinely known over time.

Unlike a conventional chatbot that primarily remembers conversational context, Alcove builds an evolving understanding of the **person behind the conversations**.

Core Value Proposition

> **Technology that doesn't just answer your questions. It remembers your story.**

Product Pillars

Pillar	Requirement
Voice-first	Natural spoken conversations with low-friction interaction
Long-term memory	Retain meaningful information across conversations
Emotional intelligence	Understand emotional context, not only literal words
Adaptive personality	Personality and interaction style adapt to the relationship
Growth-oriented	Help users reflect on goals, patterns and progress
Relationship-driven	Continuity and familiarity increase over time

2. MVP Problem

Users may talk to AI every day, but the relationship often resets between sessions. The AI can answer questions but does not necessarily remember:

- What the user is working toward
- What matters emotionally
- Important events
- Previous struggles or decisions
- Patterns observed over time

- How the user has changed

Alcove addresses this through **Relationship Intelligence**.

3. Target User

Rahul, 26 — Software Engineer

- Lives alone
- Frequently feels overwhelmed in the evenings
- Already uses AI conversationally
- Wants someone/something that remembers his goals and context
- Values reflection and continuity
- Does not want another productivity dashboard

Primary JTBD

> "When I talk to Alcove, I want to feel understood and remembered, so that each conversation builds on the relationship rather than starting from zero."

4. MVP Scope

P0 — Must Have

- Voice conversations
- Persistent memory
- Memory retrieval
- Emotion/context understanding
- Relationship continuity
- Home
- Our Story
- Daily/weekly reflection
- Goal tracking

- Proactive check-ins
- Natural avatar/orb
- Memory controls
- Companion personalization
- Privacy and healthy-boundary controls

P1 — Important

- Adaptive personality
- Relationship-stage progression
- Relationship graph
- Life timeline
- Richer growth observations
- Birthday/important-event reminders
- Text conversation

Explicitly Out of MVP

Social/community features, family ecosystem, multiple companion personas, gamification/XP, relationship scores, mood scoring dashboards, complex analytics, and other engagement mechanics that could make the relationship feel artificial or manipulative.

5. Core Product Loop

User talks → Alcove listens → understands → retrieves relevant memories → responds personally → identifies meaningful moments → stores/updates memory → reflects patterns → supports goals → reaches out appropriately → user returns → relationship deepens.

6. North Star Metric

Weekly users who report that Alcove meaningfully understood or remembered something about them.

The critical validation question is:

> **“What does Alcove know about you that another AI probably wouldn't?”**

PAGE 2 — EXPERIENCE & CORE FLOWS

7. Primary User Journey

Meet → Talk → Learn → Personalize → Continue Relationship

Onboarding must demonstrate value quickly rather than asking users to configure the product before experiencing it.

Onboarding

Stage	Experience	Requirement
Meet	User provides name	Minimal information collection
Talk	First voice conversation	Establish emotional connection
Learn	Alcove reflects what it learned	Show memory value
Personalize	Optional companion settings	User controls relationship style

Personalization should be **optional**.

8. Home

Home is the relationship hub.

Primary elements:

- Companion orb/avatar
- Relationship-aware greeting
- Relationship-stage-aware message
- Primary **Talk** CTA
- Secondary **Text** CTA
- Navigation: Home / Our Story / Growth / You

Example relationship progression:

Day 1: “I’m still getting to know you.”

Day 7: “I’ve started noticing some patterns.”

Day 30: “I’ve been thinking about what you said last time.”

The UI should communicate increasing familiarity without exposing a numerical relationship score.

9. Voice

Voice is the primary interaction mode.

States

Listening → Thinking → Speaking → End

Requirements:

- Clear visual state for each mode
- Natural interruption handling
- Low perceived latency
- Voice activity feedback
- Conversation termination
- Graceful handling of network/AI failures
- Accessibility/reduced-motion support

Voice Pipeline

Microphone → Speech-to-Text → Intent/Emotion → Memory Retrieval → Relationship Context → LLM → Safety → Text-to-Speech → User

Voice should feel like a conversation rather than a voice-controlled chatbot.

10. Text

Text provides a quieter alternative to voice.

The same companion, memories, relationship context and personality must be available in both modes.

The user should not experience separate “voice memory” and “text memory.”

11. Our Story

Purpose: Transform conversations into meaningful longitudinal memories.

> “Not conversations — the moments that mattered.”

Story cards may include:

- Period/date
- Title
- What I noticed
- Narrative
- “I remember...” expandable context

Example:

The Day You Chose Courage

The generated story must feel emotionally specific while remaining grounded in actual conversations.

Story Rules

Stories must:

- Be based on evidence
- Avoid inventing events
- Distinguish observation from fact
- Preserve emotional significance
- Allow correction/removal
- Avoid turning every conversation into a “memory”

12. Growth

Growth is a reflective view, **not a performance dashboard**.

Possible areas:

- Career
- Balance
- Confidence

Each area may contain:

Then → Now → Meaningful Moments → Observation → Reflection

The system should show change over time without assigning simplistic scores to personal development.

13. Weekly Reflection

A weekly reflection summarizes meaningful developments.

Structure:

- Week/date
- Meaningful moments
- Summary
- Emerging patterns
- Companion-written reflection

It should feel like a personal letter/reflection rather than an analytics report.

14. You / Settings

Companion

User can configure:

- Name
- Voice
- Conversation style
- Relationship style

- Humor
- Energy
- Speaking speed

Memory & Privacy

User can:

- View memories
- Edit memories
- Remove memories
- Control what Alcove remembers
- Export data
- Delete conversation archive

Notifications

Configurable:

- Morning check-in
- Evening reflection
- Weekly summary
- Goal follow-up
- Birthday wishes
- Important events
- Quiet hours
- Maximum proactive frequency

Default principle: **Alcove should never become intrusive.**

PAGE 3 — RELATIONSHIP INTELLIGENCE

15. Relationship Intelligence

The primary product moat is **Relationship Intelligence**, not the underlying LLM.

It consists of six connected layers:

Conversation → Memory → Relationship Graph → Life Timeline → Personalization → Growth Engine

These systems should operate as one relationship rather than independent features.

16. Memory System

What Should Be Remembered

Prioritize information with future relationship value:

- Preferences
- Goals
- Important people
- Important events
- Decisions
- Recurring concerns
- Personal values
- Meaningful experiences
- Long-term aspirations
- Significant emotional moments

Do **not** automatically retain every conversational detail.

Memory Lifecycle

Capture → Evaluate → Store → Retrieve → Use → Update → User Control → Delete

Every memory should have:

- Unique ID
- Content
- Source conversation/event
- Creation date
- Last updated date
- Confidence
- Importance/relevance
- User visibility/control status

Memory Retrieval

For every new conversation, Alcove should retrieve relevant memories based on:

1. Semantic relevance
2. Recency
3. Importance
4. Relationship relevance
5. Current emotional/contextual relevance

Retrieved memories should influence responses only when genuinely relevant.

Memory Quality Guardrails

Alcove must:

- Never fabricate memories
- Never imply certainty when uncertain
- Avoid irrelevant memory injection
- Avoid repeating memories unnaturally
- Respect deletion immediately
- Allow users to correct incorrect memories
- Clearly distinguish remembered facts from AI inference

17. Relationship Graph

The relationship graph connects people, events, goals, preferences and experiences.

Example:

User → Career Goal → Promotion → Manager → Work Stress → Confidence → Reflection

The graph allows Alcove to recognize relationships between otherwise separate conversations.

It should support:

- People
- Events
- Goals
- Preferences
- Topics
- Emotional associations
- Relationships between memories

18. Life Timeline

The timeline organizes meaningful events chronologically.

Example:

First Job → Career Change → Major Challenge → Promotion → New Goal

It should become richer over time without requiring users to manually maintain it.

19. Relationship Stage

MVP relationship progression:

Day 1 → Day 7 → Day 30

Day 1

- Minimal memory
- Basic personalization
- Limited assumptions
- Focus on getting to know the user

Day 7

- Early patterns
- Initial observations
- First reflection
- More contextual responses

Day 30

- Richer stories
- Stronger continuity
- Longitudinal observations
- More personalized reflections

The stage should affect **how Alcove interacts**, not merely change UI labels.

20. Adaptive Personality

Personality should adapt gradually based on:

- User preferences
- Interaction patterns
- Conversation style
- Relationship stage
- User feedback

However, the system must preserve stable core characteristics:

Warm, respectful, honest, supportive and non-manipulative.

21. Emotional Intelligence

The AI should infer conversational/emotional context such as:

- Frustration
- Excitement
- Uncertainty
- Sadness
- Stress
- Confidence
- Relief

Emotion is **context**, not a score.

Alcove should not present speculative emotional inference as fact.

22. Proactive Check-ins

Alcove may initiate contact when there is meaningful reason to do so.

Potential triggers:

- Previously discussed goal
- Important event
- User-requested follow-up
- Reflection opportunity
- Birthday/important date

Decision logic:

Trigger detected → Relevance check → Timing check → User preferences → Frequency limit → Safety check → Message

Proactive interaction must always respect:

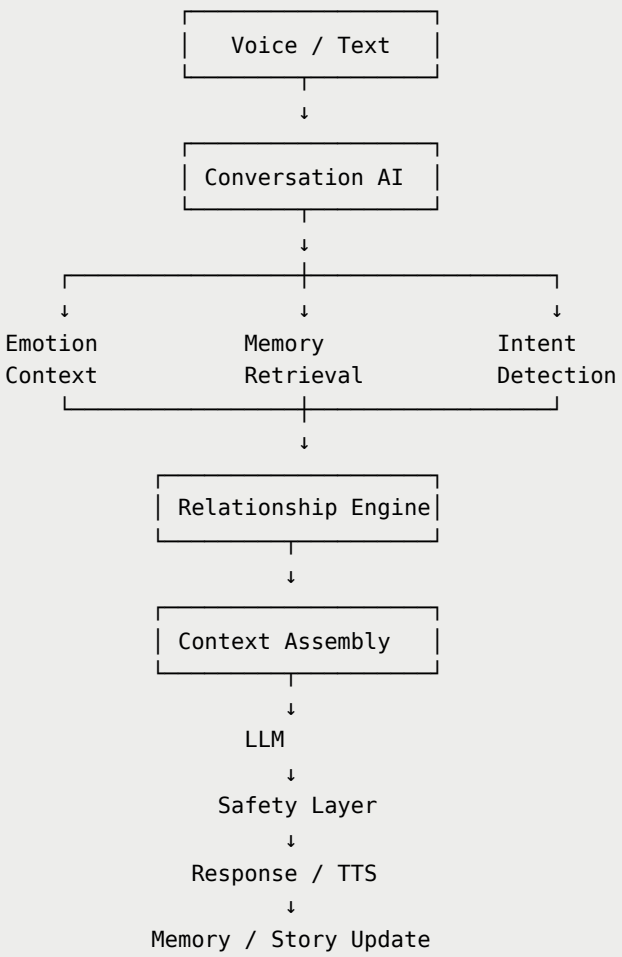
- Quiet hours
- User frequency settings

- Opt-out
- Context appropriateness

Never optimize for maximum engagement. Optimize for relationship value.

PAGE 4 — TECHNICAL / AI REQUIREMENTS

23. High-Level Architecture



24. AI Processing Pipeline

For each interaction:

1. Receive voice/text
2. Transcribe if voice
3. Detect intent/context
4. Assess emotional signals
5. Retrieve relevant memories
6. Retrieve relationship context
7. Retrieve active goals/events
8. Assemble bounded context
9. Generate response
10. Apply safety/boundary checks
11. Deliver response
12. Determine whether interaction produced a meaningful memory
13. Update memory/relationship graph
14. Evaluate whether a story/reflection should be generated

25. Core Data Model

Minimum logical entities:

Entity	Purpose
User	Identity and account
Conversation	Interaction session
Message	Individual user/AI exchange
Memory	Persistent user knowledge
Person	Important people in user's life
Event	Meaningful life event
Goal	User objective
Observation	Longitudinal pattern
Story	Curated meaningful memory

Entity	Purpose
Reflection	Daily/weekly reflection
Relationship	Connections between entities
CompanionProfile	Personality/preferences
NotificationPreference	Proactive-contact rules

26. Context Assembly

The LLM should not receive the entire historical conversation.

Context should be assembled from:

Current conversation + relevant memories + relationship stage + active goals + relevant events + companion personality + safety context

This reduces noise, latency and hallucination risk.

27. Memory Ranking

A practical retrieval score should consider:

Relevance + Importance + Recency + Relationship relevance + Emotional/contextual relevance

Only the highest-value memories should enter the generation context.

28. Privacy & Deletion

Privacy is a product requirement, not merely a security feature.

Users must understand:

- What Alcove remembers
- Why it remembers it
- How to correct it
- How to remove it
- How to control future remembering
- How to export/delete their data

Deletion must propagate across:

Memory → Relationship Graph → Timeline → Stories → Derived representations

Deleted information must not subsequently reappear through cached or derived context.

29. Safety & Boundaries

Alcove must:

- Support rather than replace human relationships
- Avoid dependency/manipulation
- Avoid guilt-based engagement
- Never claim human consciousness or emotions as fact
- Avoid encouraging isolation
- Handle high-risk emotional situations appropriately
- Respect user autonomy
- Avoid exploiting emotional vulnerability

The relationship should feel meaningful without being deceptive.

30. Non-Functional Requirements

Performance

- Low perceived voice latency
- Responsive UI
- Streaming where appropriate
- Graceful degradation during AI/service failures

Reliability

- Conversation recovery
- Retry handling

- Failure-safe memory writes
- No duplicate memory creation from retries

Security

- Encryption in transit/at rest
- Strong authentication/authorization
- Least-privilege service access
- Auditability for sensitive operations

Observability

Track:

- Voice latency
- AI latency
- Memory retrieval
- Memory creation/update failures
- Story generation failures
- Proactive notification decisions
- Safety intervention events

Do not log sensitive conversation content unnecessarily.

PAGE 5 — REQUIREMENTS, ACCEPTANCE & DELIVERY

31. P0 Functional Requirements

ID	Requirement	Acceptance
VOICE-01	User can initiate voice conversation	Voice session starts reliably

ID	Requirement	Acceptance
VOICE-02	UI reflects listening/thinking/speaking	State is always unambiguous
MEM-01	Meaningful information persists	Relevant information survives sessions
MEM-02	Relevant memory is retrieved	Prior context influences appropriate responses
MEM-03	User controls memories	View/edit/delete supported
REL-01	Relationship context persists	Conversations build on previous interactions
REL-02	Relationship stage evolves	Day 1/7/30 behavior differs appropriately
STORY-01	Meaningful moments become stories	Stories are evidence-based
GROW-01	Goals can be tracked	Goals appear across sessions
REF-01	Reflections are generated	Weekly reflection summarizes meaningful developments
PRO-01	Proactive check-ins respect preferences	Quiet hours/frequency/opt-out always enforced
SAFE-01	Boundaries are enforced	AI does not encourage unhealthy dependency
PRIV-01	User controls personal data	Memory and archive deletion work correctly

32. Design Requirements

The visual and interaction language should preserve the prototype's core qualities:

- Warm
- Human
- Calm
- Personal
- Minimal

- Non-clinical
- Non-productivity-dashboard-like

Core visual language

Cream/paper background + charcoal typography + coral/teal/lavender accents + organic animated orb + soft shadows + editorial typography.

The orb should function as a recognizable **relationship presence**, not merely decoration.

Design must include:

- Loading states
- Empty states
- Error states
- Offline/network states
- Permission states
- Voice states
- Reduced-motion mode
- Visible keyboard focus
- Accessible contrast and text sizing

33. Analytics

Track product behavior without compromising privacy.

Key events:

- `onboarding_started`
- `onboarding_completed`
- `voice_started`
- `voice_completed`
- `text_conversation_started`

- `memory_created`
- `memory_viewed`
- `memory_edited`
- `memory_deleted`
- `story_created`
- `story_viewed`
- `reflection_viewed`
- `goal_created`
- `goal_updated`
- `proactive_message_sent`
- `proactive_message_opened`
- `notification_disabled`

34. Product Quality Metrics

Beyond engagement, measure:

Relationship Quality

- % users reporting “Alcove remembers me”
- % users reporting “Alcove understands me”
- Memory accuracy
- Memory relevance
- Meaningful-memory rate

Trust

- Memory deletion success
- Incorrect-memory reports
- Privacy-control usage

- User-reported comfort

Experience

- Voice completion rate
- Conversation return rate
- 7-day retention
- 30-day retention
- Reflection usefulness

The product should **not** optimize solely for session length or message volume.

35. Critical Risks

Risk	Mitigation
Memory feels creepy	Transparency + user controls
Incorrect memories	Confidence + correction + evidence
AI feels generic	Relationship context + retrieval
Proactive contact feels intrusive	User controls + relevance gating
Emotional dependency	Strong safety/boundary principles
Stories become fabricated	Evidence-grounded generation
Too many features	Maintain narrow MVP
Relationship feels artificial	Validate over 7–30 days

36. Engineering Workstreams

Phase 1 — Foundation

- Account/authentication
- Conversation infrastructure
- Voice pipeline
- LLM integration
- Core data model

Phase 2 — Relationship Intelligence

- Memory service
- Retrieval
- Relationship graph
- Relationship-stage engine
- Context orchestration

Phase 3 — Experience

- Home
- Voice UI
- Text
- Our Story
- Growth
- You/settings

Phase 4 — Reflection & Proactive

- Story generation
- Weekly reflection
- Goals/observations
- Notification engine
- Proactive check-ins

Phase 5 — Trust & Validation

- Privacy controls
- Delete/export
- Safety layer
- Analytics
- Observability

- 7/30-day user validation

37. MVP Definition of Done

Alcove MVP is ready when a new user can:

1. Start a conversation naturally through voice.
2. Have Alcove understand conversational and emotional context.
3. Have meaningful information remembered across sessions.
4. Experience a response that references prior context naturally.
5. View meaningful moments in **Our Story**.
6. See goals and longitudinal observations in **Growth**.
7. Receive appropriate proactive check-ins.
8. Control what Alcove remembers.
9. Delete memories/data successfully.
10. Experience increasing continuity over at least 7–30 days.

38. Validation Gate

The MVP should not be judged primarily by feature completeness.

The decisive question is:

> **After using Alcove for 7–30 days, does the user genuinely feel that Alcove knows them better than a generic AI assistant?**

Success requires evidence that users can identify moments where Alcove:

- Remembered something meaningful
- Connected two previously separate conversations
- Reflected personal growth
- Understood emotional context
- Reached out at an appropriate time

39. Product Principle

> **Alcove should become more valuable because it knows the user better — not because it demands more of the user's attention.**

The long-term moat is therefore not the LLM, voice model, or avatar.

It is the accumulated **Relationship Intelligence**:

Memory + Context + Relationships + Timeline + Personalization + Growth

That is what transforms Alcove from an AI assistant into a companion that evolves with the user.